# Research Report

2026-03-16

## Introduction

Electric vehicle (EV) prices vary substantially across models, even among vehicles with similar technical specifications. Battery capacity is one of the most visible EV characteristics and is often associated with both production cost and consumer value. However, vehicle price is not determined by battery size alone. Manufacturers differ in brand positioning, and premium brands may charge systematically higher prices than non-premium brands even for vehicles with similar specifications.

This report studies EV models listed for sale in Germany and investigates whether battery capacity is associated with vehicle price, and whether that association differs between premium and non-premium manufacturers. The research questions are:

1. Among electric vehicle models listed for sale in Germany, is battery capacity positively associated with vehicle price?
2. Does the association between battery capacity and price differ between premium and non-premium manufacturers?

The data come from the *Electric Vehicle Specifications and Prices* dataset obtained from Kaggle and compiled by web scraping EV Database. EV Database aggregates manufacturer-reported technical specifications and listed prices for EVs sold in Germany. The dataset contains 360 observations, where each row represents a distinct EV model. Since the data are observational rather than experimental, all conclusions in this report are associative rather than causal.

The response variable is `Price.DE.`, the listed vehicle price in euros. The main explanatory variable is `Battery`, measured in kilowatt-hours (kWh). A second key variable is manufacturer segment, coded as premium or non-premium. Following the proposal, Tesla, BMW, Audi, Mercedes-Benz, and Porsche are classified as premium brands. Other quantitative variables that may be useful during model building include `Range`, `Efficiency`, `Top_speed`, `Acceleration..0.100.`, and `Fast_charge`.

This report uses linear regression because the response is continuous and the main goal is inference about average relationships. In particular, including an interaction between battery capacity and market segment allows the battery-price relationship to differ between premium and non-premium manufacturers, which directly addresses the second research question. Based on the course material, variables named in the research question remain in the model throughout model selection even if other selection criteria would remove them.

## Data Import and Preparation

The raw dataset is stored locally in the project folder as `data/EV_cars.csv`. Using a relative file path makes the analysis reproducible for all group members and avoids machine-specific paths such as `~/Desktop/...`.

```r
library(dplyr)
library(ggplot2)
library(knitr)
```

```r
data_path <- "data/EV_cars.csv"
ev_raw <- read.csv(data_path, header = TRUE, na.strings = c("NA", ""))
```

```r
names(ev_raw)
```

```
## [1] "Battery"              "Car_name"             "Car_name_link"
## [4] "Efficiency"           "Fast_charge"          "Price.DE."
## [7] "Range"                "Top_speed"            "acceleration..0.100."
```

```r
dim(ev_raw)
```

```
## [1] 360   9
```

```r
head(ev_raw)
```

```
##   Battery                            Car_name
## 1    75.0 Tesla Model Y Long Range Dual Motor
## 2    57.5                       Tesla Model 3
## 3    60.5                          BYD ATTO 3
## 4    61.7               MG MG4 Electric 64 kWh
## 5    75.0 Tesla Model 3 Long Range Dual Motor
## 6    57.5                       Tesla Model Y
##                                                          Car_name_link
## 1 https://ev-database.org/car/1619/Tesla-Model-Y-Long-Range-Dual-Motor
## 2                      https://ev-database.org/car/1991/Tesla-Model-3
## 3                         https://ev-database.org/car/1782/BYD-ATTO-3
## 4              https://ev-database.org/car/1708/MG-MG4-Electric-64-kWh
## 5 https://ev-database.org/car/1992/Tesla-Model-3-Long-Range-Dual-Motor
## 6                      https://ev-database.org/car/1743/Tesla-Model-Y
##   Efficiency Fast_charge Price.DE. Range Top_speed acceleration..0.100.
## 1        172         670     59017   435       217                  5.0
## 2        137         700     46220   420       201                  6.1
## 3        183         370     44625   330       160                  7.3
## 4        171         630     39990   360       160                  7.9
## 5        149         780     55220   505       201                  4.4
## 6        164         580     47567   350       217                  6.9
```

The raw file contains 360 observations and 9 recorded variables: `Battery`, `Car_name`, `Car_name_link`, `Efficiency`, `Fast_charge`, `Price.DE.`, `Range`, `Top_speed`, and `acceleration..0.100.`. The response variable `Price.DE.` has 51 missing values, and `Fast_charge` has 2 missing values. Since price is the response, rows with missing price cannot be used for the regression analysis and are removed. The two rows with missing `Fast_charge` are retained during initial cleaning and are only excluded in models that include `Fast_charge`.

The next chunk creates a manufacturer variable from the vehicle name, constructs the premium/non-premium grouping from the proposal, renames the acceleration variable to a cleaner column name, and keeps the variables relevant to the analysis. Identifier variables such as `Car_name_link` are not used as predictors because they do not have an interpretable statistical relationship with price.

```r
premium_brands <- c("Tesla", "BMW", "Audi", "Mercedes-Benz", "Mercedes", "Porsche")

ev <- ev_raw %>%
  mutate(
    manufacturer = sub(" .*", "", Car_name),
    manufacturer = ifelse(manufacturer == "Mercedes-Benz", "Mercedes-Benz", manufacturer),
    market_segment = ifelse(manufacturer %in% premium_brands, "Premium", "Non-premium"),
    market_segment = factor(market_segment, levels = c("Non-premium", "Premium")),
    Acceleration_0_100 = `acceleration..0.100.`
  ) %>%
  select(
```

```r
    Car_name,
    Car_name_link,
    manufacturer,
    market_segment,
    Price.DE.,
    Battery,
    Range,
    Efficiency,
    Top_speed,
    Acceleration_0_100,
    Fast_charge
  ) %>%
  filter(
    !is.na(Price.DE.),
    !is.na(Battery),
    !is.na(market_segment)
  )

summary(ev)
```

```
##    Car_name         Car_name_link      manufacturer           market_segment
##  Length:309         Length:309         Length:309         Non-premium:215
##  Class :character   Class :character   Class :character   Premium    : 94
##  Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##    Price.DE.         Battery           Range         Efficiency
##  Min.   : 22550   Min.   : 21.30   Min.   :135.0   Min.   :137.0
##  1st Qu.: 45690   1st Qu.: 57.40   1st Qu.:295.0   1st Qu.:171.0
##  Median : 56942   Median : 70.50   Median :380.0   Median :187.0
##  Mean   : 67264   Mean   : 71.08   Mean   :369.1   Mean   :195.4
##  3rd Qu.: 73100   3rd Qu.: 85.00   3rd Qu.:445.0   3rd Qu.:209.0
##  Max.   :218000   Max.   :123.00   Max.   :685.0   Max.   :295.0
##
##    Top_speed     Acceleration_0_100  Fast_charge
##  Min.   :125.0   Min.   : 2.100     Min.   : 170.0
##  1st Qu.:155.0   1st Qu.: 4.800     1st Qu.: 335.0
##  Median :180.0   Median : 6.700     Median : 520.0
##  Mean   :181.1   Mean   : 7.308     Mean   : 552.8
##  3rd Qu.:200.0   3rd Qu.: 9.000     3rd Qu.: 680.0
##  Max.   :282.0   Max.   :19.100     Max.   :1290.0
##                                     NA's   :2
```

```r
nrow(ev)
```

```
## [1] 309
```

After removing rows with missing price, the main analysis dataset contains 309 EV models. A quick check of duplicate `Car_name` values shows that some names appear more than once. These rows were retained because they are not exact duplicates of the full record: for example, duplicate names can have different `Car_name_link` values and different charging specifications, suggesting separate listings or updated versions rather than accidental repetition.

```r
ev_raw %>%
  summarise(
    n_raw = n(),
    missing_price = sum(is.na(Price.DE.)),
    missing_fast_charge = sum(is.na(Fast_charge))
  )
```

```
##   n_raw missing_price missing_fast_charge
## 1   360            51                   2
```

```r
ev %>%
  count(market_segment)
```

```
##   market_segment   n
## 1    Non-premium 215
## 2        Premium  94
```

```r
ev %>%
  count(Car_name) %>%
  filter(n > 1) %>%
  arrange(desc(n), Car_name)
```

```
##         Car_name n
## 1 Mini Cooper SE 2
## 2  Tesla Model Y 2
```

For the main regression, one practical choice is to fit models using complete cases on the variables included in each model. If `Fast_charge` remains in a candidate model, the effective sample size becomes 307 rather than 309 because of the two missing charging values. This should be noted when comparing models.

## Exploratory Analysis

Exploratory analysis serves two purposes in this study. First, it helps describe the data to readers who are not familiar with the EV market. Second, it helps determine whether the assumptions of a linear regression model are plausible and whether any transformations or model simplifications should be considered.

Start with summary statistics for the response and candidate explanatory variables.

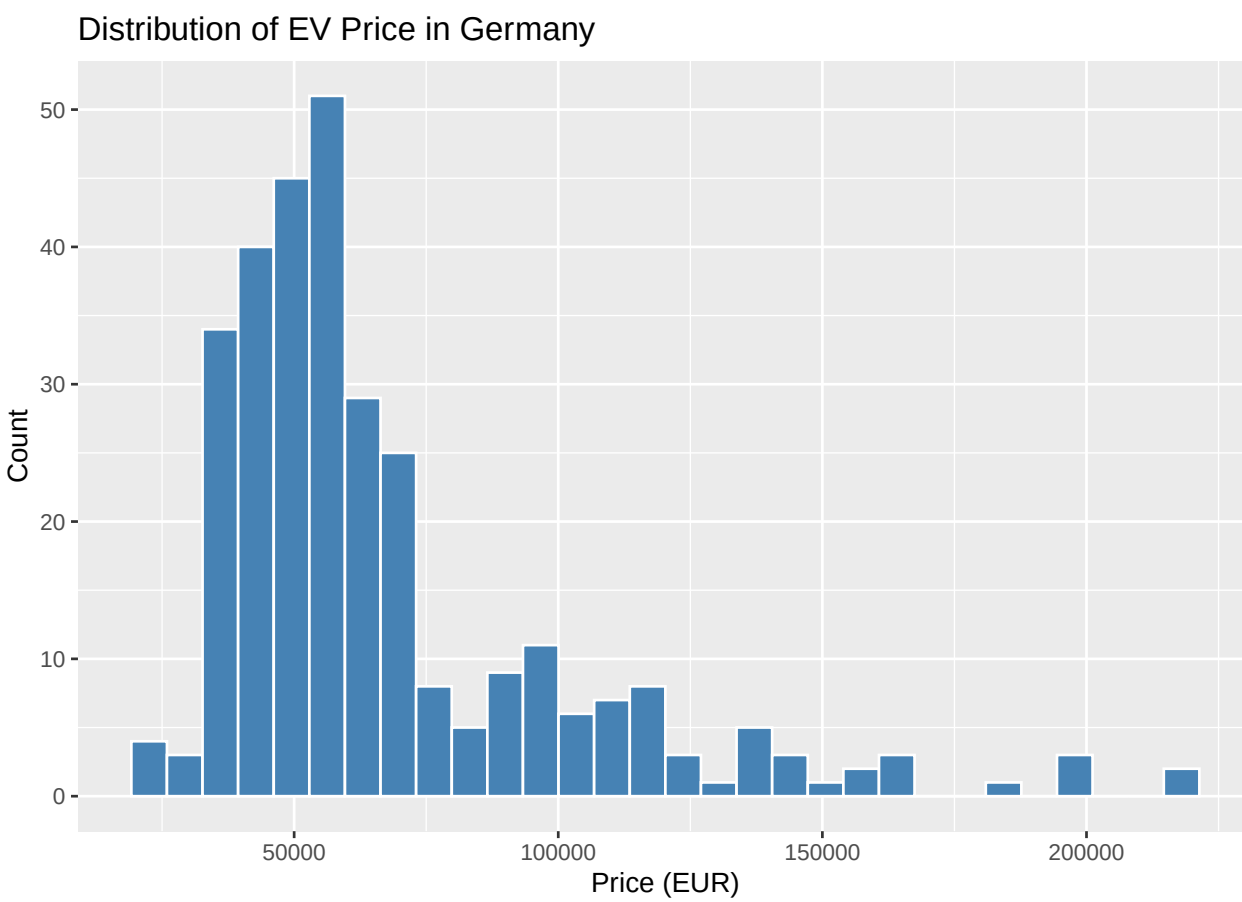
```r
summary_table <- ev %>%
  summarise(
    n = n(),
    mean_price = mean(Price.DE., na.rm = TRUE),
    median_price = median(Price.DE., na.rm = TRUE),
    sd_price = sd(Price.DE., na.rm = TRUE),
    mean_battery = mean(Battery, na.rm = TRUE),
    median_battery = median(Battery, na.rm = TRUE),
    sd_battery = sd(Battery, na.rm = TRUE),
    mean_range = mean(Range, na.rm = TRUE),
    mean_efficiency = mean(Efficiency, na.rm = TRUE)
  )

kable(summary_table, digits = 2)
```

| n | mean_price | median_price | sd_price | mean_battery | median_battery | sd_battery | mean_range | mean_efficiency |
|---|---|---|---|---|---|---|---|---|
| 309 | 67264.28 | 56942 | 34508.25 | 71.08 | 70.5 | 20.65 | 369.13 | 195.41 |

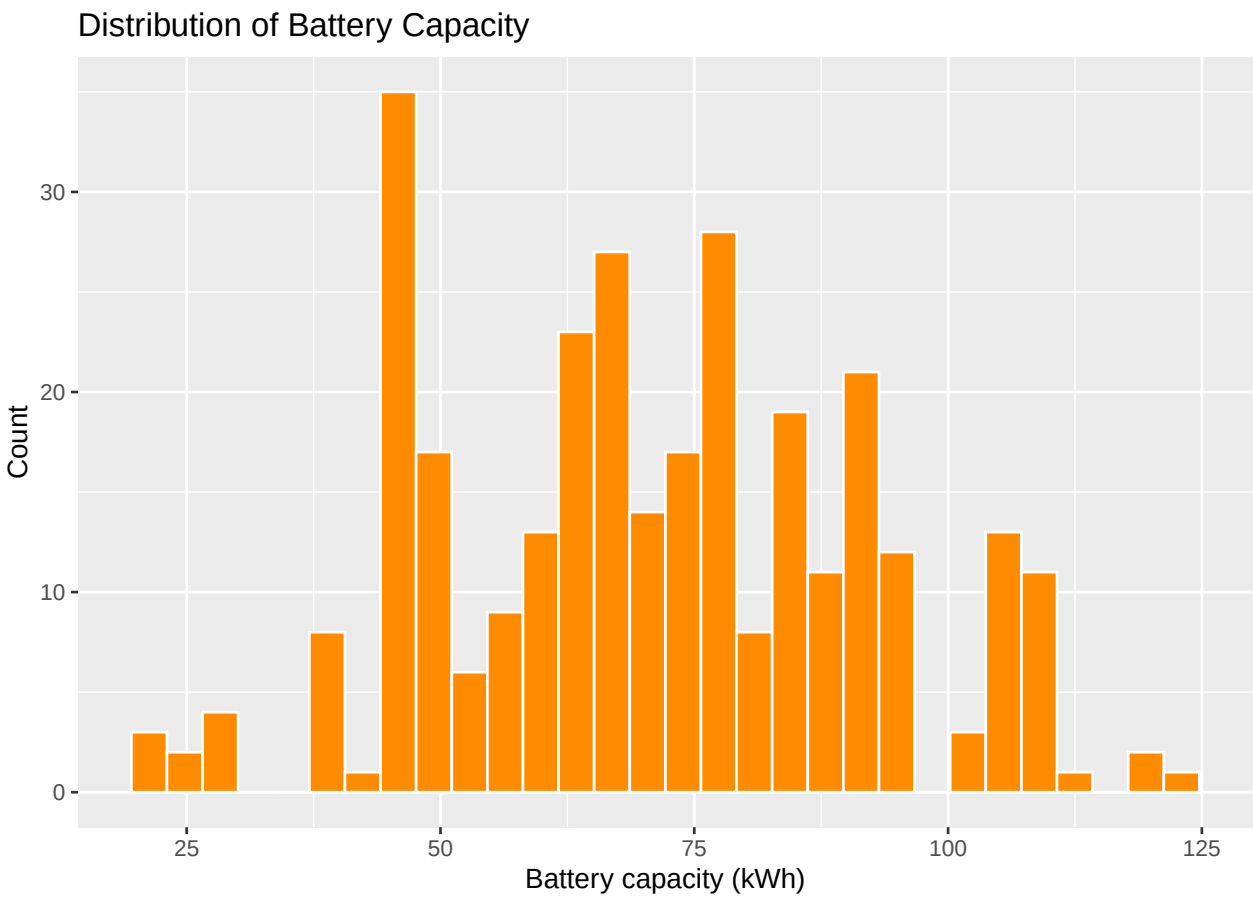
Among the 309 EV models used in the main analysis, the mean listed price is about 67,264 EUR and the median price is 56,942 EUR, indicating some right-skew from a smaller number of high-priced vehicles. Battery capacity has a mean of 71.1 kWh and a median of 70.5 kWh, so the battery distribution is more balanced than the price distribution. When the sample is split by manufacturer segment, premium vehicles are clearly larger and more expensive on average: premium models have a mean price of about 94,359 EUR and a mean battery size of 84.0 kWh, compared with 55,418 EUR and 65.4 kWh for non-premium models.

The distribution of price should be examined because strong right-skew or a few very expensive vehicles could affect both model fit and interpretation. The distribution of battery capacity should also be assessed to understand the range of the main predictor.

```
ggplot(ev, aes(x = Price.DE.)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(
    title = "Distribution of EV Price in Germany",
    x = "Price (EUR)",
    y = "Count"
  )
```



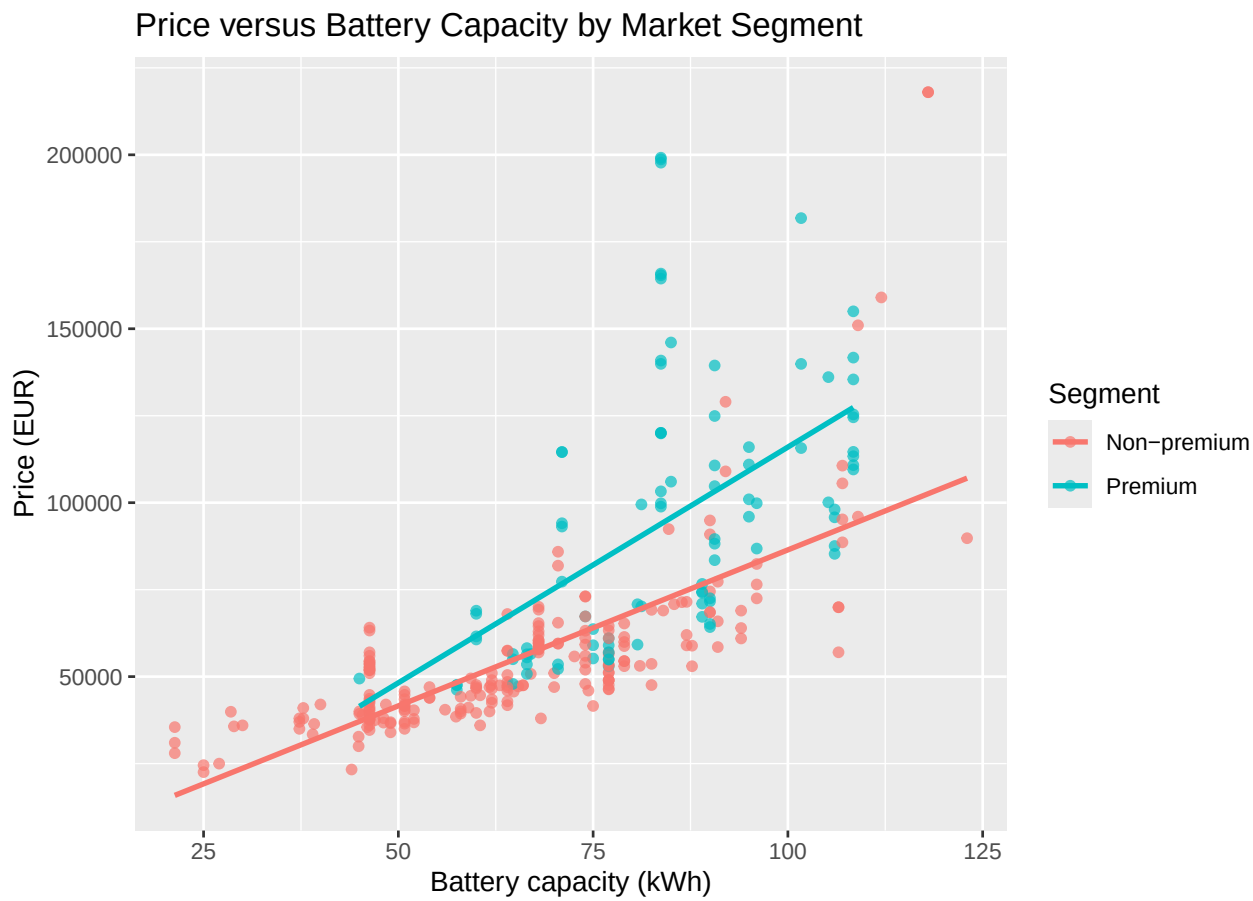

```
ggplot(ev, aes(x = Battery)) +
  geom_histogram(bins = 30, fill = "darkorange", color = "white") +
  labs(
    title = "Distribution of Battery Capacity",
    x = "Battery capacity (kWh)",
    y = "Count"
  )
```




The price histogram shows noticeable right-skew, which is expected in a market where a small number of luxury and high-performance vehicles are much more expensive than the rest of the sample. The battery histogram is more concentrated, with many vehicles clustered in the mid-size battery range and fewer models at the smallest and largest battery capacities.

The most important exploratory plot is the scatterplot of price against battery capacity, with separate fitted lines for premium and non-premium manufacturers. This directly addresses both research questions.

```
ggplot(ev, aes(x = Battery, y = Price.DE., color = market_segment)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Price versus Battery Capacity by Market Segment",
    x = "Battery capacity (kWh)",
    y = "Price (EUR)",
    color = "Segment"
  )
```




The scatterplot indicates a clear positive association between battery capacity and price overall. It also shows strong separation between the two market segments. Premium vehicles tend to appear both higher and farther to the right on the plot, meaning that they generally have larger batteries and higher prices. Simple fitted lines within each segment are positive, with an estimated unadjusted increase of about 897 EUR per kWh for non-premium vehicles and about 1,355 EUR per kWh for premium vehicles. This visual pattern suggests that the battery-price relationship may indeed differ by manufacturer segment.

Because several quantitative vehicle characteristics may be related to one another, it is also useful to examine pairwise relationships and correlations. This can help identify potential multicollinearity before model fitting.

```
quant_vars <- ev %>%
  select(
    Price.DE.,
    Battery,
    Range,
    Efficiency,
    Top_speed,
    Acceleration_0_100,
    Fast_charge
  )

round(cor(quant_vars, use = "complete.obs"), 2)
```

```
##                    Price.DE. Battery Range Efficiency Top_speed
## Price.DE.               1.00    0.70  0.59       0.17      0.76
## Battery                 0.70    1.00  0.88       0.16      0.70
## Range                   0.59    0.88  1.00      -0.31      0.76
```

```
## Efficiency              0.17    0.16 -0.31       1.00     -0.21
## Top_speed               0.76    0.70  0.76      -0.21      1.00
## Acceleration_0_100     -0.53   -0.61 -0.75       0.42     -0.84
## Fast_charge             0.62    0.59  0.72      -0.29      0.79
##                    Acceleration_0_100 Fast_charge
## Price.DE.                       -0.53        0.62
## Battery                         -0.61        0.59
## Range                           -0.75        0.72
## Efficiency                       0.42       -0.29
## Top_speed                       -0.84        0.79
## Acceleration_0_100               1.00       -0.68
## Fast_charge                     -0.68        1.00
```

The correlation matrix also highlights why model selection is needed. Price is strongly positively correlated with `Battery` (0.70) and `Top_speed` (0.76), while `Battery` and `Range` are themselves highly correlated (0.88). `Top_speed` and `Acceleration_0_100` are also strongly associated in magnitude, with a correlation of -0.84 because faster vehicles have lower acceleration times. These patterns suggest substantial overlap among the technical variables, so a simpler regression model is preferable for interpretation.

# Statistical Analysis

## Modeling Strategy

The response is continuous, so a linear regression model is appropriate. The variables appearing explicitly in the research question, `Battery`, `market_segment`, and their interaction, remain in all candidate models. Additional quantitative variables are treated as optional controls and may be removed during model selection if they do not meaningfully improve the model.

Following the course guidance for inference-focused projects, test-based or criteria-based model selection should be used rather than pure prediction-based selection. Here, the primary selection approach is backward elimination using partial F-tests on optional variables only, while keeping the research-question terms fixed in the model. AIC and BIC are then reported as supporting criteria for comparing candidate models.

## Full Model

```r
full_model <- lm(
  Price.DE. ~ Battery * market_segment +
    Range + Efficiency + Top_speed + Acceleration_0_100 + Fast_charge,
  data = ev
)

summary(full_model)
```

```
##
## Call:
## lm(formula = Price.DE. ~ Battery * market_segment + Range + Efficiency +
##     Top_speed + Acceleration_0_100 + Fast_charge, data = ev)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -47909 -10469   -368   7603  94516
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   -1.597e+05  2.406e+04  -6.640 1.48e-10 ***
```

```
## Battery                       -3.146e+02  3.597e+02  -0.874 0.382574
## market_segmentPremium         -4.454e+04  1.185e+04  -3.757 0.000206 ***
## Range                          8.162e+01  6.666e+01   1.225 0.221718
## Efficiency                     4.270e+02  1.140e+02   3.746 0.000215 ***
## Top_speed                      6.427e+02  7.239e+01   8.879  < 2e-16 ***
## Acceleration_0_100             1.034e+03  7.666e+02   1.349 0.178505
## Fast_charge                    1.686e+01  7.556e+00   2.232 0.026373 *
## Battery:market_segmentPremium  6.276e+02  1.429e+02   4.391 1.57e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17870 on 298 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.7383, Adjusted R-squared:  0.7313
## F-statistic: 105.1 on 8 and 298 DF,  p-value: < 2.2e-16
```

The coefficient of `Battery` estimates the battery-price association for the baseline group, which is non-premium manufacturers. The coefficient of `market_segmentPremium` estimates the difference in intercept between premium and non-premium manufacturers. The interaction term `Battery:market_segmentPremium` estimates how much the battery-price slope differs for premium brands relative to non-premium brands.

In the full model, the interaction term is positive and highly statistically significant (estimate about 628, p-value $< 0.001$), indicating that the battery-price association differs by segment even after controlling for the other vehicle characteristics. `Top_speed`, `Efficiency`, and `Fast_charge` are also statistically significant, while `Range` and `Acceleration_0_100` contribute less once the other variables are present. The full model has an adjusted $R^2$ of about 0.731, so it explains a substantial proportion of the variation in listed price.

## Backward Selection with Required Terms Retained

The following function performs backward elimination using `drop1(..., test = "F")`. It only considers removing optional variables. The three research-question terms are always retained.

```r
backward_keep_required <- function(data, response, required_terms, optional_terms, alpha = 0.05) {
  current_optional <- optional_terms

  repeat {
    current_terms <- c(required_terms, current_optional)
    current_formula <- as.formula(
      paste(response, "~", paste(current_terms, collapse = " + "))
    )
    current_model <- lm(current_formula, data = data)

    if (length(current_optional) == 0) {
      break
    }

    drop_table <- drop1(current_model, test = "F")
    removable <- intersect(rownames(drop_table), current_optional)

    if (length(removable) == 0) {
      break
    }

    removable_table <- drop_table[removable, , drop = FALSE]
    worst_term <- rownames(removable_table)[which.max(removable_table$`Pr(>F)`)]
```

```r
    worst_p <- max(removable_table$`Pr(>F)`, na.rm = TRUE)

    if (is.na(worst_p) || worst_p <= alpha) {
      break
    }

    current_optional <- setdiff(current_optional, worst_term)
  }

  final_terms <- c(required_terms, current_optional)
  final_formula <- as.formula(
    paste(response, "~", paste(final_terms, collapse = " + "))
  )

  list(
    final_model = lm(final_formula, data = data),
    kept_optional_terms = current_optional,
    removed_optional_terms = setdiff(optional_terms, current_optional)
  )
}
```

```r
required_terms <- c("Battery", "market_segment", "Battery:market_segment")
optional_terms <- c("Range", "Efficiency", "Top_speed", "Acceleration_0_100", "Fast_charge")

selection_results <- backward_keep_required(
  data = ev,
  response = "Price.DE.",
  required_terms = required_terms,
  optional_terms = optional_terms,
  alpha = 0.05
)

selection_results$kept_optional_terms
```

```
## [1] "Efficiency"  "Top_speed"   "Fast_charge"
```

```r
selection_results$removed_optional_terms
```

```
## [1] "Range"              "Acceleration_0_100"
```

```r
final_model <- selection_results$final_model
summary(final_model)
```

```
##
## Call:
## lm(formula = final_formula, data = data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -47707 -10548   -710   7534 101699
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   -1.193e+05  1.050e+04 -11.361  < 2e-16 ***
## Battery                        6.344e+01  8.685e+01   0.730 0.465706
```

```
## market_segmentPremium         -4.122e+04  1.150e+04  -3.583 0.000396 ***
## Efficiency                     3.359e+02  3.787e+01   8.872  < 2e-16 ***
## Top_speed                      5.654e+02  5.604e+01  10.090  < 2e-16 ***
## Fast_charge                    2.027e+01  7.344e+00   2.760 0.006136 **
## Battery:market_segmentPremium  5.980e+02  1.383e+02   4.324 2.09e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17930 on 300 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.7347, Adjusted R-squared:  0.7294
## F-statistic: 138.4 on 6 and 300 DF,  p-value: < 2.2e-16
```

Backward selection removed `Range` and `Acceleration_0_100`, while retaining `Efficiency`, `Top_speed`, and `Fast_charge` in addition to the required research-question terms. This matches the exploratory evidence of multicollinearity: once `Battery`, `Top_speed`, and the other retained controls are in the model, `Range` no longer provides much additional information. The selected model keeps the key interaction term and remains highly explanatory, with an adjusted $R^2$ of about 0.729.

## Criteria-Based Comparison of Candidate Models

Although the main model-selection argument should remain inference-focused, AIC and BIC can still be used as supporting evidence when comparing a small number of sensible candidate models.

```r
model_battery_only <- lm(Price.DE. ~ Battery * market_segment, data = ev)
model_selected <- final_model
model_full <- full_model

data.frame(
  model = c("Battery + segment + interaction", "Selected model", "Full model"),
  AIC = c(AIC(model_battery_only), AIC(model_selected), AIC(model_full)),
  BIC = c(BIC(model_battery_only), BIC(model_selected), BIC(model_full)),
  adj_R2 = c(
    summary(model_battery_only)$adj.r.squared,
    summary(model_selected)$adj.r.squared,
    summary(model_full)$adj.r.squared
  )
)
```

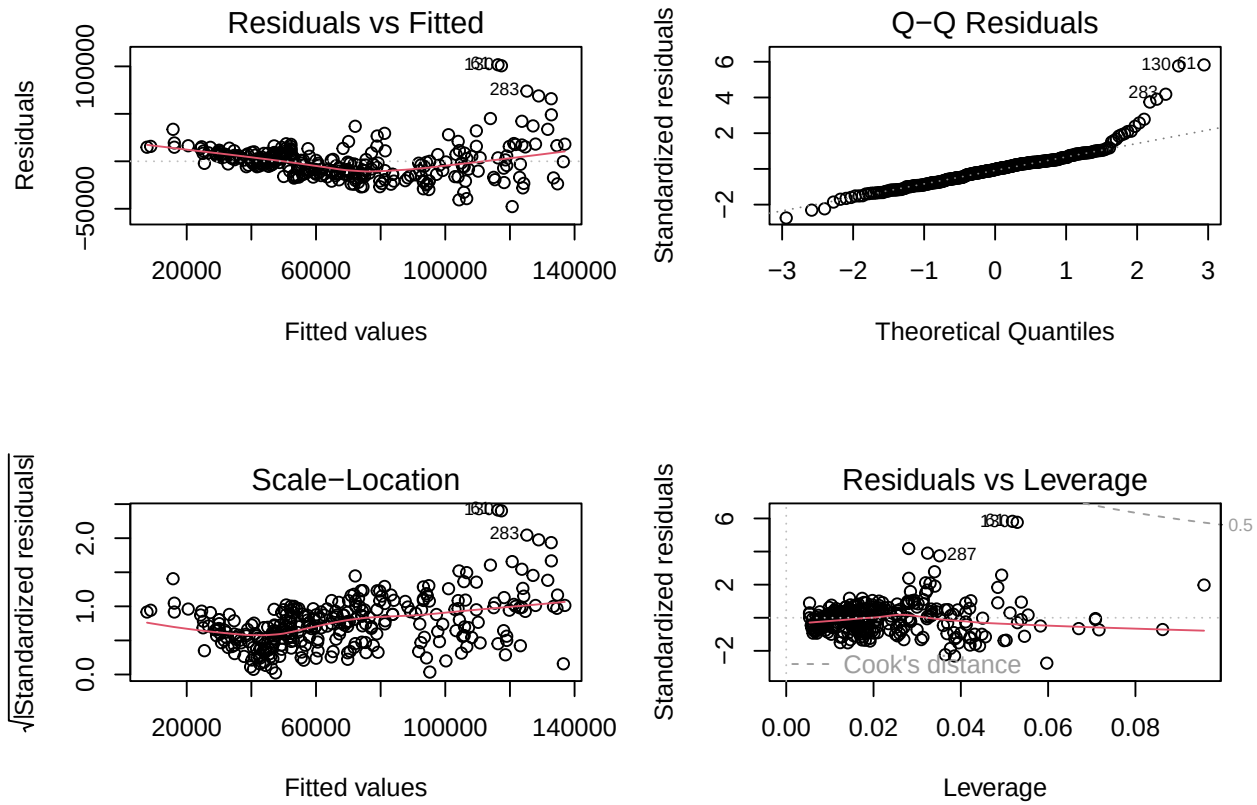
```
##                             model      AIC      BIC    adj_R2
## 1 Battery + segment + interaction 7083.748 7102.415 0.5641101
## 2                  Selected model 6893.754 6923.569 0.7293555
## 3                      Full model 6893.529 6930.797 0.7312636
```

The model comparison supports using the selected model. The battery-plus-segment-only model has a much lower adjusted $R^2$ of about 0.564, so the additional controls clearly improve fit. The selected and full models have almost identical AIC values (6893.75 versus 6893.53) and very similar adjusted $R^2$ values (0.729 versus 0.731). Because the full model improves fit only negligibly while using two additional variables, the selected model is the more parsimonious choice.

## Regression Diagnostics

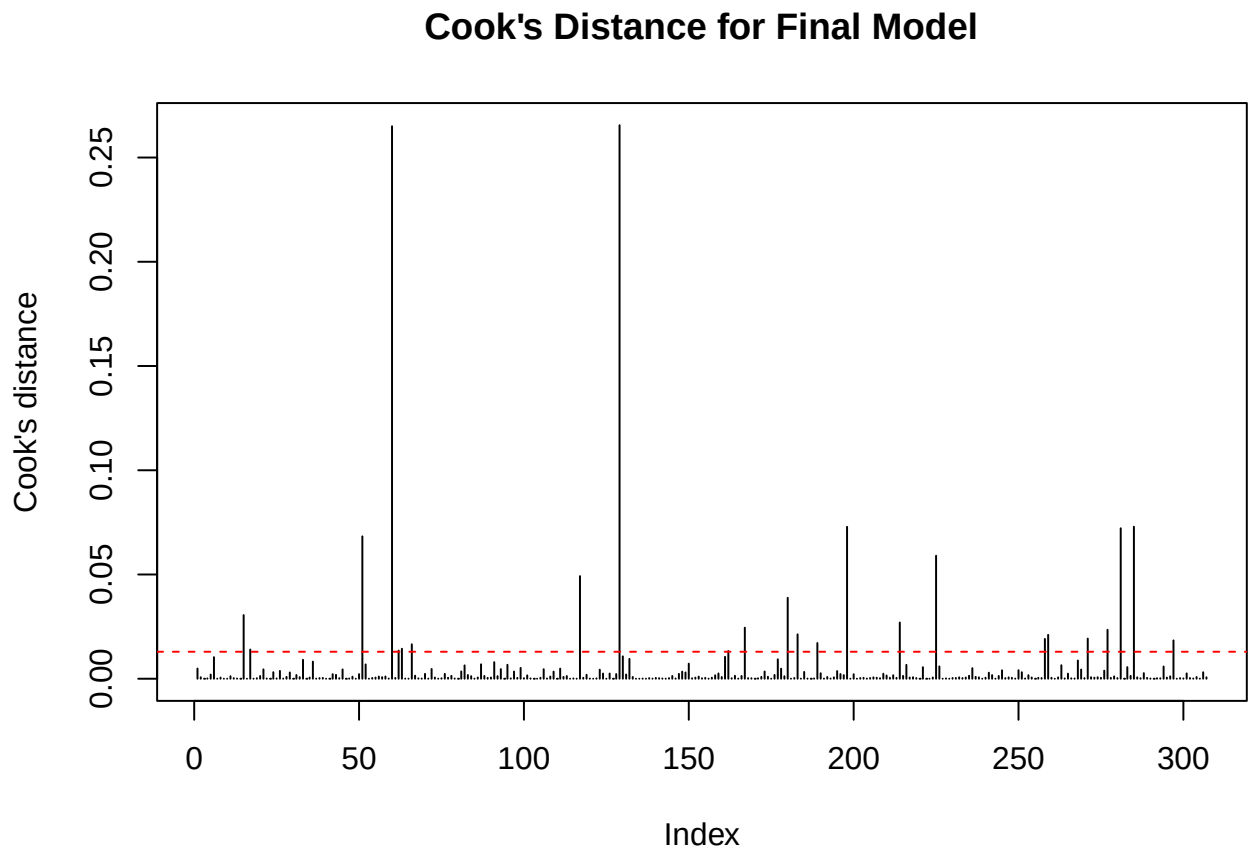
After selecting the final model, standard diagnostic checks should be performed. These checks help assess linearity, constant variance, approximate normality of residuals, and influential observations.

```r
par(mfrow = c(2, 2))
plot(final_model)
```




```r
par(mfrow = c(1, 1))
```

```r
plot(cooks.distance(final_model), type = "h",
     main = "Cook's Distance for Final Model",
     ylab = "Cook's distance")
abline(h = 4 / nrow(ev), col = "red", lty = 2)
```



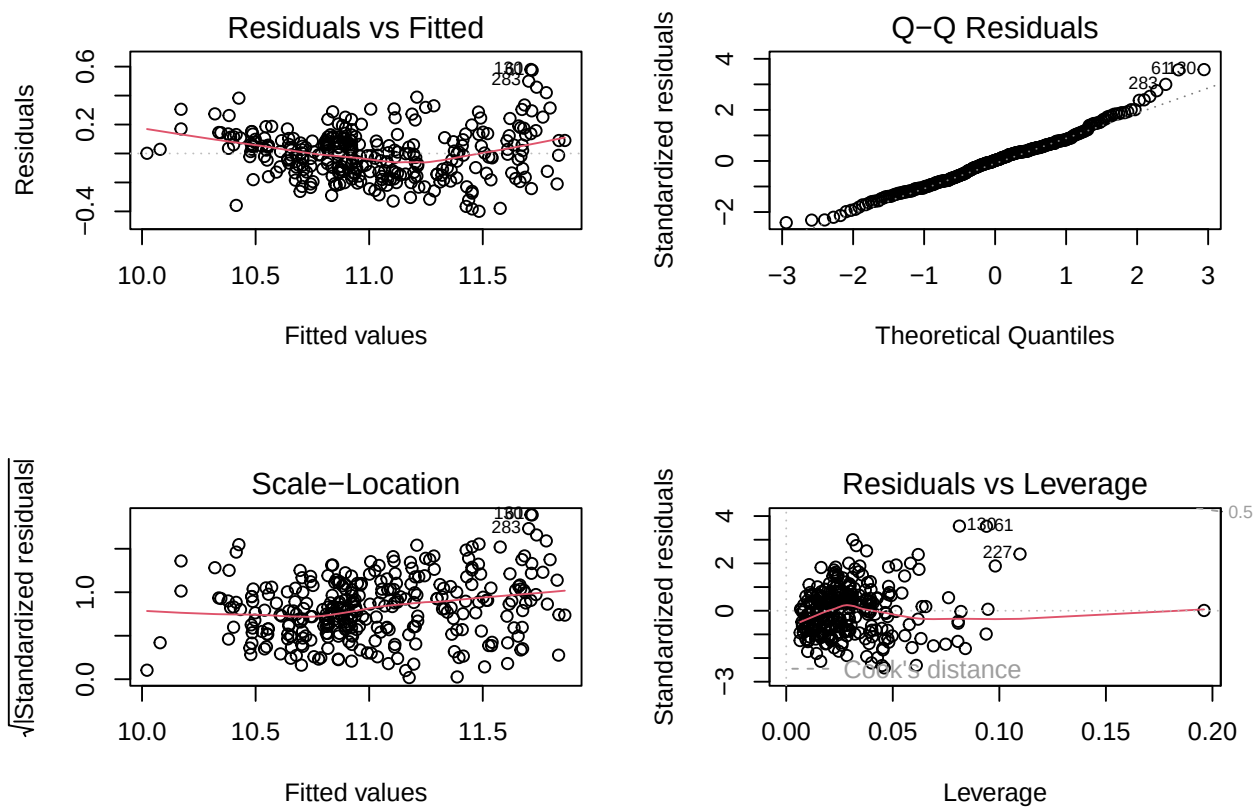

If the residual plots show strong heteroscedasticity or right-skewed residuals, a log transformation of price may be worth considering. The transformed model should only be adopted if diagnostics clearly improve and the interpretation remains appropriate for the study objective.

```
log_model <- lm(
  log(Price.DE.) ~ Battery * market_segment +
    Range + Efficiency + Top_speed + Acceleration_0_100 + Fast_charge,
  data = ev
)

summary(log_model)
```

```
##
## Call:
## lm(formula = log(Price.DE.) ~ Battery * market_segment + Range +
##     Efficiency + Top_speed + Acceleration_0_100 + Fast_charge,
##     data = ev)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.39920 -0.12827  0.00337  0.09458  0.58027
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   8.077e+00  2.279e-01  35.447  < 2e-16 ***
## Battery                      -5.621e-03  3.407e-03  -1.650 0.100012
## market_segmentPremium        -3.078e-01  1.123e-01  -2.741 0.006487 **
## Range                         1.900e-03  6.313e-04   3.010 0.002837 **
```

```
## Efficiency                     7.874e-03  1.080e-03   7.294 2.73e-12 ***
## Top_speed                      5.296e-03  6.856e-04   7.724 1.73e-13 ***
## Acceleration_0_100            -7.139e-03  7.260e-03  -0.983 0.326291
## Fast_charge                    2.742e-04  7.156e-05   3.832 0.000155 ***
## Battery:market_segmentPremium  5.234e-03  1.354e-03   3.866 0.000136 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1692 on 298 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.8453, Adjusted R-squared:  0.8412
## F-statistic: 203.6 on 8 and 298 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(2, 2))
plot(log_model)
```




```r
par(mfrow = c(1, 1))
```

The log-price sensitivity analysis fits somewhat better, with an adjusted $R^2$ of about 0.841, which suggests that a transformation may help stabilize residual variation. Importantly, the central substantive result remains the same in the log model: the battery-by-segment interaction is still positive and highly significant. This makes the main conclusion more robust, even if the report ultimately presents the simpler untransformed model for ease of interpretation.

## Reporting the Final Model

Once the final model is chosen, report the coefficient table and interpret the coefficients in the context of the research questions.

```r
kable(round(summary(final_model)$coefficients, 4),
      caption = "Coefficient estimates for the final regression model")
```

Table 2: Coefficient estimates for the final regression model

| | Estimate | Std. Error | t value | Pr(>\|t\|) |
|---|---|---|---|---|
| (Intercept) | -119258.2125 | 10496.8459 | -11.3613 | 0.0000 |
| Battery | 63.4378 | 86.8515 | 0.7304 | 0.4657 |
| market_segmentPremium | -41215.8559 | 11502.2868 | -3.5833 | 0.0004 |
| Efficiency | 335.9446 | 37.8674 | 8.8716 | 0.0000 |
| Top_speed | 565.3966 | 56.0359 | 10.0899 | 0.0000 |
| Fast_charge | 20.2689 | 7.3438 | 2.7600 | 0.0061 |
| Battery:market_segmentPremium | 597.9874 | 138.2935 | 4.3240 | 0.0000 |

In the selected model, the estimated coefficient for `Battery` in the non-premium group is about 63 EUR per kWh and is not statistically significant on its own ($p = 0.466$). However, the interaction term is about 598 EUR per kWh and highly significant ($p < 0.001$), so the implied battery slope for premium vehicles is much steeper, about 661 EUR per kWh. This provides strong evidence that the battery-price relationship differs between premium and non-premium manufacturers after controlling for the retained technical variables. The coefficient on `market_segmentPremium` by itself is negative, but because that term is defined at a battery size of zero, it is not substantively meaningful on its own. The more relevant interpretation comes from the interaction and the fitted lines: premium vehicles are priced differently and their prices rise more steeply with battery capacity.

# Discussion and Conclusion

The exploratory plots and the regression analysis both point to the same broad conclusion: EV price is positively associated with battery capacity, but the strength of that association depends on manufacturer segment. In the raw scatterplot, both segments show upward trends, and premium vehicles tend to occupy the higher-price, larger-battery portion of the figure. After controlling for efficiency, top speed, and fast-charging capability, the selected regression model shows that the battery-price slope is much steeper for premium brands than for non-premium brands.

For the first research question, the evidence supports a positive overall association between battery capacity and vehicle price among EVs listed in Germany. For the second research question, the interaction term provides strong evidence that this relationship differs by segment. In the selected model, premium vehicles gain substantially more expected price per additional kWh than non-premium vehicles. This suggests that battery capacity is valued differently across market segments, consistent with premium brands using larger batteries as part of a broader high-end product strategy.

The retained control variables also help clarify the pricing structure. `Top_speed`, `Efficiency`, and `Fast_charge` remain statistically important in the selected model, indicating that price reflects a bundle of performance and charging characteristics rather than battery size alone. By contrast, `Range` and `Acceleration_0_100` do not provide enough additional information once the retained variables are included, which is consistent with the strong correlations observed during the exploratory analysis.

Several limitations remain important. The data are observational and based on public listings, so the results should not be interpreted causally. The premium/non-premium classification is a simplified representation of brand positioning, and some important determinants of price, such as trim level, software features, interior quality, and brand-specific technology, are not directly measured. In addition, the analysis is specific to vehicles listed in Germany and may not generalize to other markets.

Overall, the analysis supports the conclusion that battery capacity is related to EV price, but not in the

same way for all manufacturers. Premium brands appear to attach a stronger price premium to additional battery capacity, and this difference remains visible even after accounting for other technical characteristics.

# Statement of AI Use

Generative AI was used to help organize this report draft and convert the project plan into a reproducible markdown document with R code for data cleaning, exploratory analysis, regression modeling, model selection, and diagnostics. The statistical conclusions, numerical results, and model interpretations should be based only on the output produced from the actual dataset after the code is run.

# Appendix

## Notes on Course Alignment

The modeling strategy in this report is aligned with the course guidance from the provided lecture material:

- Interaction terms are used because the research question asks whether the battery-price relationship differs by manufacturer segment.
- Model selection is treated as an inference problem, so the main procedure uses test-based or criteria-based comparison rather than purely predictive methods.
- Variables named directly in the research question remain in the model throughout selection.